



AMENI KHABTHANI

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SUMMARY

I am a Data Scientist with a Master's in Data science, AI, and HealthTech, seeking opportunities to apply my expertise to real-world challenges. I am passionate about developing innovative models; I thrive in collaborative environments and excel at solving complex challenges. I aim to deliver impactful, data-driven solutions that make a meaningful difference.

EXPERIENCE

Data Science Intern

Satoripop, Sousse

6 months - 2024

- Built ASR and TTS solutions, enabling accurate speech processing in French, English, and Tunisian dialects.
- Fine-tuned Whisper model for Speech Recognition, reducing the word error rate (WER) by 45% on the target dataset
- Fine-tuned XTTS v2 model for Speech Synthesis, improving speech naturalness and intelligibility based on user feedback.
- Built APIs using the FastAPI framework, decreasing response time by 30% and ensuring seamless integration with external applications.
- Designed user-friendly interfaces for APIs with the Gradio toolkit, improving accessibility and usability.

Data Science Intern

Stunas, Sousse

2 months - 2023

- Developed an OCR model for invoice label recognition using DocTr for text detection and NER techniques for label extraction, reducing manual processing time by 50%.

ACADEMIC PROJECTS

ECG Segmentation

- Heartbeat classification using CNN architecture.

Brain MRI Tumor Detection

- Improved the accuracy and efficiency of brain tumor detection by 85% using CNN.

Sentiment Analysis and Rating Prediction for Medicines

- Created an ML model that forecasts medicine ratings based on various features, leveraging sentiment analysis on user reviews.

Length of Stay Prediction for Hospital Management

- Established a machine-learning model to predict patient length of stay in healthcare facilities, leveraging historical patient data and clinical features to enhance resource allocation and optimize hospital management.
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FREELANCE PROJECTS

Creating a Custom TTS System

- Development of a Text-to-Speech (TTS) model based on XTTS v2 fine-tuned from scratch on Maghrebi voice data.
- Assist and support the client in implementing the chain of the solution: data preparation, training, model evaluation.

AI-based Astrology API (End-to-End)

- Design and deployment of an intelligent astrology API combining traditional and modern sources.
- Integration of an LLM with prompt engineering, intent classification, and generation of personalized responses.
- Database management with Supabase for storing user profiles, query histories, and custom configurations.

Seamless Dubbing Automation

- Replaced the existing ElevenLabs-based dubbing service in an API with a custom multilingual dubbing pipeline.
- The new module includes speaker diarization and leverages Whisper Large v3 for ASR, GPT-4 for translation, and hybrid methodology XTTS + Orpheus TTS for TTS synthesis.
- Integrated into a Flask API hosted on Azure App Service
- Handled speech-to-text, speaker segmentation, translation, synthesis, and audio merging to produce aligned, natural-sounding dubbed content.

[In Progress] Medical ASR & Text-to-Translation System

- Currently developing an ASR system for Modern Standard Arabic (MSA) that preserves domain-specific medical terminology.
- Parallel development of a text-to-translation (T2TT) pipeline for Arabic-to-English translation, emphasizing clinical and semantic accuracy.
- Designed for future integration into healthcare transcription and multilingual medical communication tools.

EDUCATION

MASTER'S DEGREE

Pristini School of AI Oct - 2022 / Oct-2024

- Data science, AI, and HealthTech

BACHELOR'S DEGREE

Iset Mahdia Sep - 2019 / Jun-2022

- Electrical Engineering Specializing in Automation and Industrial Computing.

CERTIFICATIONS

- Natural Language Processing Specialization [\(Link\)](#)
 - Generative AI with Large Language Models [\(Link\)](#)
 - Python data structures [\(Link\)](#)
 - Deep learning AI Specialization [\(Link\)](#)
- Generative AI Engineering with LLM Specialization [\(Link\)](#)
 - AWS Academy Cloud Foundations [\(Link\)](#)
 - Data Analysis with Python [\(Link\)](#)

SKILLS

- Machine Learning • Deep Learning • Scikit-learn • Tensorflow • Pytorch • Transformer • LLM
- Generative AI • TTS • STT • Python • DAX • Gradio • FastAPI • Matplotlib • Seaborn • SQL • NOSQL
- Hadoop • Hive • Communication • Problem-Solving

LANGUAGES

INVOLVEMENT

PASSIONS

Arabic (native)	Board team member in Pristini Robotics club	Swimming
English (Professional)	Board team member in Arsii Pristini club	ART activities
French (Professional)		
Italian (Notions)		